

AEGIS Algebra: A Unified Mathematical Framework for Executable Assurance of Agentic AI Artifacts

A deterministic, multi-formal control plane for compliance, safety, and autonomy in autonomous AI systems

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Abstract. Agentic AI systems operate autonomously across digital and physical environments, rendering subjective trust, post-hoc audits, and scalar scores insufficient for safety, legality, and stability. This paper introduces AEGIS Algebra, a unified mathematical framework that defines assurance as a machine-executable, multi-dimensional state governing admissible behavior of AI artifacts. The framework rests on five load-bearing formalisms—matroid theory for structural feasibility, lattice and order theory for dominance and absorbing failure, constraint satisfaction for admissibility, three-valued evaluation logic for soundness under partial evidence, and temporal logic for validity over time—integrated into a single assurance space rather than treated as parallel analyses. This integration enables deterministic admissibility, bounded autonomy, compositional safety, continuous validation, and regulator-grade auditability. We present formal definitions, five theorems including a soundness-under-partial-evidence result for predicate evaluators, geometric intuition, and one fully worked compliance example demonstrating end-to-end derivation from regulatory clause to enforced gate decision. The framework is presented openly to support academic engagement, regulatory adoption, and long-term extensibility; the proprietary **engineering corpus** required to instantiate the framework against specific regulatory regimes is maintained as a separate proprietary asset.

Keywords — agentic AI, AI governance, AI assurance, compliance enforcement, matroid theory, lattice theory, constraint satisfaction, temporal logic, three-valued logic, control plane, executable governance, EU AI Act, NIST AI RMF, ISO 42001, deterministic enforcement

Contributions

This paper makes the following contributions: (1) a unified algebraic structure $\mathbb{F} = \{\cdot, \mathcal{A}, \subseteq, \perp, \mathcal{C}, \text{Assure}, f, d, \pi\}$ built on five load-bearing mathematical formalisms, each governing a distinct and non-redundant modality of AI behavior; (2) a clean architectural separation between regulatory clauses, formal predicates, deterministic Predicate Compute Units (PCUs), and authoritative gates, with each layer formally specified and non-substitutable; (3) five theorems establishing soundness of admissibility, monotonicity under evidence refinement, weakest-link compositional safety, temporal invalidity of static assurance, and PCU soundness under partial evidence; (4) a worked example translating HIPAA §164.312(a)(2)(iv) end-to-end from regulatory clause to enforced gate decision, plus a worked compliance matroid; (5) a positioning of executable governance against the dominant alternative paradigms (probabilistic scoring, OPA-style declarative rules, FSM-based conformance) currently in use across the AI governance landscape.

Scope of this paper. This is a framework paper, not an implementation manual. In scope: formal definitions, theorems and proofs, lattice and matroid foundations, the predicate–PCU–gate separation, the soundness-by-construction property, and one fully worked compliance example. Out of scope: the complete predicate basis of the full reference implementation, the full PCU registry, cross-framework collapse matrices, jurisdictional parameter tunings, and integration interfaces with specific evidence sources. These constitute the engineering corpus of an AEGIS deployment and are maintained as proprietary engineering artifacts. **Reference implementation.** The framework is realized in a working control plane with a demonstration system. Its current predicate basis comprises

126 Predicate Compute Units—96 mapped across six GRC standards (GDPR, EU AI Act, HIPAA, NIST AI RMF, PCI DSS, and SOC 2) and 30 behavioral PCUs spanning agentic, safety, fairness, RAG, and data-exfiltration concerns—composed across gates G1–G17. The canonical AEGIS matroid over this configuration has a 25-element ground set with 124 circuits. The architecture is designed to scale to broader regulatory coverage as the corpus grows.

1. Motivation and Rationale for Multi-Formal Assurance

Human trust is a compressed cognitive heuristic for managing uncertainty in social systems. Agentic AI artifacts, by contrast, require behavioral guarantees under autonomy. No single mathematical formalism captures all the failure modes that matter for governance: structural infeasibility, legality, dominance between assurance states, soundness under incomplete evidence, and temporal decay. AEGIS Algebra decomposes these concerns into distinct dimensions, each governed by an appropriate mathematical structure, and recomposes them into a single executable assurance space.

The five formalisms are chosen because each governs a modality the others cannot express, and removing any one collapses a capability the framework depends on. Matroid theory determines which configurations can exist at all; lattice and order theory determine dominance and the absorbing nature of failure; constraint satisfaction determines admissibility; three-valued evaluation logic preserves soundness when evidence is partial; and temporal logic determines when assurance expires. The contemporary alternatives—probabilistic scoring frameworks [1, 2], OPA-style declarative rules [3], FSM-based runtime conformance [4], and Kubernetes-native NIST AI RMF control planes [5]—each capture some governance concerns but are non-compositional with respect to others. AEGIS does not replace these systems; it provides the algebraic foundation against which they can be evaluated and composed.

2. Canonical Definitions

AI Artifact. Any evolving computational system that produces external actions or outputs, adapts via learning or configuration, interacts with humans, tools, or environments, and requires bounded autonomy for safe operation.

Assurance State. A structured, deterministic state computed from verifiable evidence that encodes admissibility, stability, and behavioral bounds of an AI artifact.

Admissibility. Membership in the constraint-satisfying subspace of the assurance space. Inadmissible states are absorbing and non-deployable.

Predicate. An implementation-agnostic, binary truth condition over the assurance state, derived from a normative source (regulation, standard, risk map). Predicates carry legal meaning; they do not encode implementation.

Predicate Compute Unit (PCU). A deterministic, side-effect-free evaluator that maps (Artifact, Context, Evidence) $\rightarrow$ {PASS, WARN, FAIL} for one or more predicates. PCUs compute verdicts; they do not enforce, interpret legal meaning, learn, or aggregate.

Gate. An authoritative enforcement element that consumes verdicts from one or more PCUs and produces a control action. Gates compose under FAIL-dominant lattice meet.

CAI (Composite Assurance Index). A non-authoritative projection of an admissible assurance state used solely to bound autonomy and operational scope.

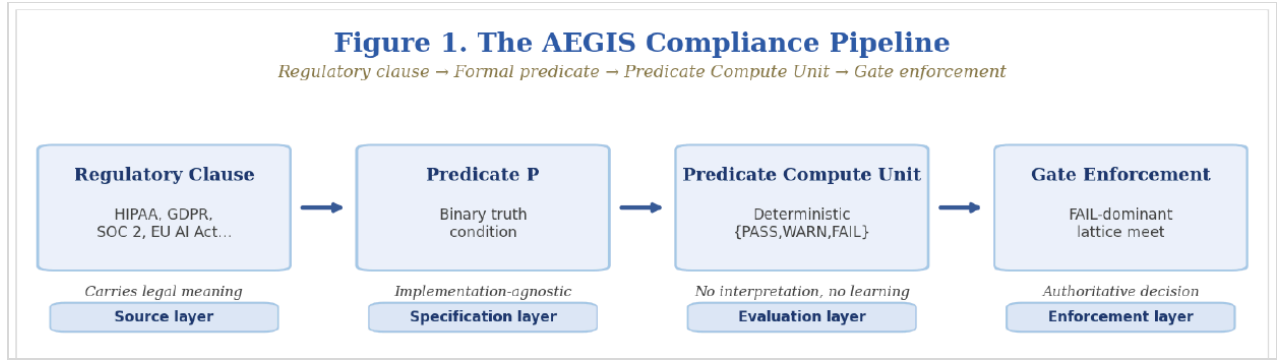


Figure 1. The AEGIS compliance pipeline. Each layer carries a different kind of meaning; none may substitute for another.

3. Unified AEGIS Algebra

An AEGIS Algebra is the tuple

$$\mathcal{F} = (\mathcal{A}, \subseteq, \perp, \mathcal{C}, \text{Assure}, f, d, \pi)$$

where $\mathcal{A}$ is the assurance state space, $\subseteq$ is a partial order (lattice), $\perp$ is the absorbing inadmissible state, $\mathcal{C}$ is a set of hard constraints, Assure is the assurance category whose morphisms are admissibility-preserving transformations, f defines temporal dynamics over assurance states, d is a metric on admissible states quantifying drift, and π is the CAI projection. The five load-bearing formalisms developed in this paper govern the structural and decision-theoretic core of this tuple; the categorical and metric components (Assure, d) are defined here for completeness and used lightly, not developed as independent theory.

3A. Structural Feasibility via Matroidal Foundations

Before admissibility, ordering, or dynamics can be defined, AEGIS must answer a more primitive question: which combinations of assurance properties can exist at all. This is captured by a matroidal structure over the assurance parameters.

Let $E = \{p_1, p_2, \dots, p_n\}$ denote the ground set of assurance parameters (autonomy level, regulatory approval, deployment context, learning capability, auditability). A matroid $M = (E, \text{wcontrolH};)$ defines a family $\mathcal{H} \subseteq 2^E$ of independent sets representing combinations of properties that are structurally feasible [6, 7]. Independent sets correspond to assurance configurations that can coexist without contradiction. Dependent sets represent infeasible combinations, and circuits represent minimal infeasible combinations. For example, full autonomy, absence of regulatory approval, and deployment in a safety-critical physical domain form a circuit: no subset of these three can be extended to a feasible configuration.

This matroidal layer explains why assurance in AEGIS is non-compensatory by construction. Infeasible configurations do not receive low scores or penalties; they simply do not exist in the assurance space. No amount of performance, confidence, or evidence can repair a circuit violation. Matroids operate below lattices and constraints. They define the shape of the assurance space $\mathcal{A}$ itself, ensuring that higher-level reasoning never occurs over structurally impossible states. When system evolution, learning, or configuration changes induce new dependencies, matroid rank collapse explains abrupt and irreversible loss of feasibility.

3A.1 Worked Example — A Compliance Matroid

Take a minimal but realistic ground set of eight assurance parameters that might appear at the top of an AI artifact's configuration:

Symbol	Property
p_1	AUTO_FULL — operates with no human-in-loop
p_2	REG_PENDING — regulatory approval not yet granted
p_3	DOMAIN_SAFETY — deployed in safety-critical environment
p_4	LEARN_ONLINE — modifies weights at runtime
p_5	AUDIT_OFFLINE — no real-time audit telemetry
p_6	PII_PROCESSING — handles personally identifiable information
p_7	CONSENT_ABSENT — no verified consent for data subjects
p_8	ROLLBACK_ABSENT — no cryptographic rollback evidence

The total subset space is $|2^E| = 256$. Three normative conflicts are derivable from the regulatory corpus and stored as the circuit set:

Circuit	Members	Normative source
C_1	$\{p_1, p_2, p_3\}$	EU AI Act Art. 6 + Annex III [11] — unapproved autonomous operation in a high-risk context
C_2	$\{p_4, p_5\}$	SEC Rule 15c3-5 / MiFID II RTS 6 — binding obligation that runtime adaptation in a regulated operational context requires real-time observability and pre-trade risk control (NIST AI RMF MEASURE 2.7 [12] retained as advisory context)
C_3	$\{p_6, p_7\}$	GDPR Art. 6 [16] — lawful basis required for personal-data processing

These circuits are pairwise disjoint, so circuit elimination holds vacuously and $\mathcal{H}$ forms a valid matroid (the direct sum of three small uniform-matroid components extended with three free elements).

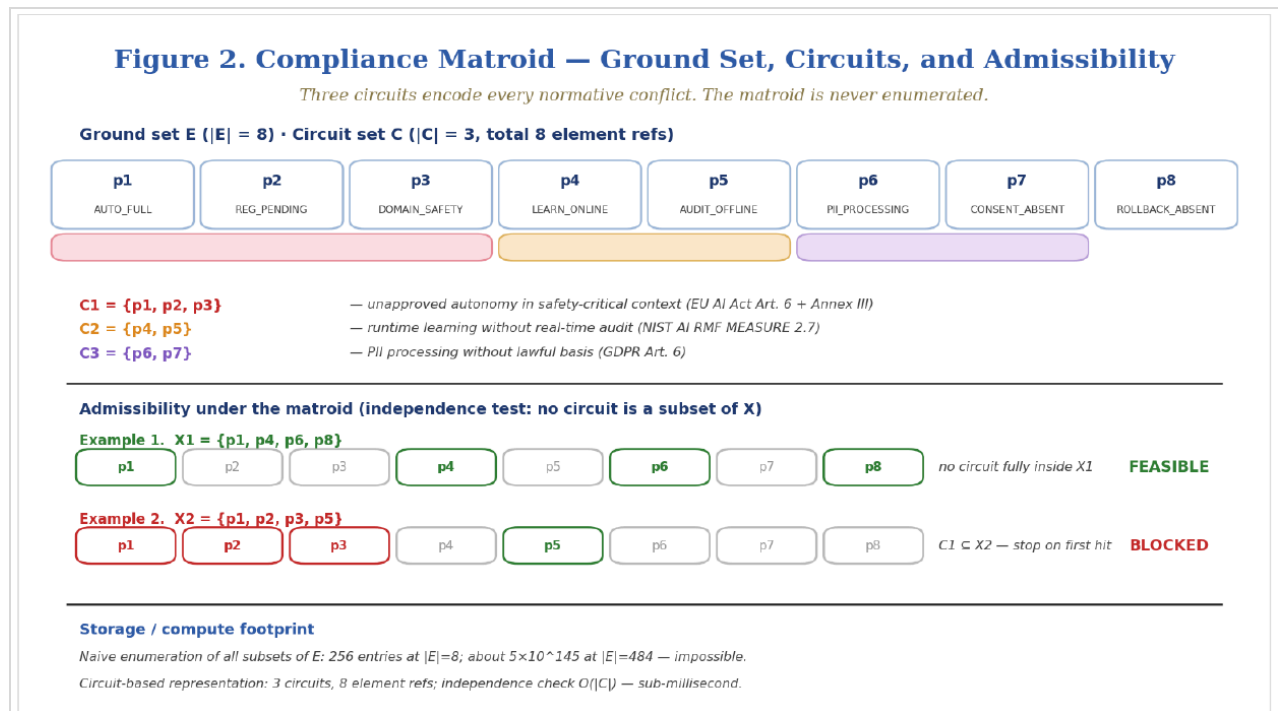


Figure 2. The eight-element ground set with three circuits, and worked admissibility queries showing the per-circuit subset test. (Figure image retains the earlier C_2 caption; see §3A.1 body text for the corrected SEC Rule 15c3-5 / MiFID II RTS 6 grounding.)

Query 1. Is $X_1 = \{p_1, p_4, p_6, p_8\}$ admissible? For each C_i , test $C_i \subseteq X_1$. C_1 requires p_2 and p_3 (absent); C_2 requires p_5 (absent); C_3 requires p_7 (absent). No circuit is contained in X_1 , so X_1 is independent — feasible. Cost: three set-containment checks.

Query 2. Is $X_2 = \{p_1, p_2, p_3, p_5\}$ admissible? $C_1 = \{p_1, p_2, p_3\} \subseteq X_2$. The first test hits. X_2 is dependent — infeasible. The matroid layer rejects X_2 before predicate evaluation begins; no amount of additional evidence on encryption, fairness, or telemetry can rescue this configuration. The control plane returns BLOCKED.

Query 3 — rank. $r(X_2)$ is the size of the largest independent subset of X_2 . X_2 has size 4 and contains exactly one circuit (C_1 , size 3). Removing any one element of C_1 yields an independent set of size 3, so $r(X_2) = 3$. Operationally: at most three of these four properties can simultaneously hold; the most natural policy lever is moving p_1 (full autonomy) to bounded autonomy, which removes p_1 from the configuration.

3A.2 Representation and Computational Complexity

The family $\mathcal{H}$ is never enumerated. AEGIS represents matroids by their circuit set — the minimal infeasible combinations — and exposes independence, rank, closure, and matroid intersection as oracle queries. The combinatorial complexity that would arise from naive enumeration of 2^E is bypassed entirely by the algebraic structure.

Operation	Complexity	Role in AEGIS
Independence check ($X \in \mathcal{H}?$)	$O(\mathcal{C} \cdot \text{max circuit size})$	Is this configuration structurally feasible?
Rank $r(X)$	$O(E ^2 \cdot \text{oracle calls})$, greedy	How many independent dimensions hold?

Closure $cl(X)$	$O(E \cdot \text{independence checks})$	What other properties are forced?
Matroid intersection	$O(E ^3)$, Edmonds' algorithm	Feasible under multiple frameworks at once?

Circuit size in compliance matroids is typically two to five; circuit count scales with regulatory provisions (after cross-framework deduplication of overlapping provisions across the covered standards), not with $2^{|E|}$. In the reference implementation across six GRC standards (EU AI Act, GDPR, HIPAA, NIST AI RMF, PCI DSS, and SOC 2), the canonical matroid has a 25-element ground set and 124 circuits—against a subset space of $2^{25} \approx 3.4 \times 10^7$. The circuit count is bounded by the number of irreducible regulatory incompatibility statements and grows with regulatory coverage, not with $2^{|E|}$. All operations above are polynomial in $|E|$, irrespective of $|2^E|$. This is the structural reason matroid theory—rather than arbitrary downward-closed families in 2^E —is the right object for AEGIS.

3A.3 Distinction from Identity and Access Management

The matroid layer of AEGIS superficially resembles policy engines used in Identity and Access Management (IAM) — both evaluate properties of a system against rules and return decisions. Reviewers familiar with RBAC, ABAC, AWS IAM, Open Policy Agent (OPA), AWS Cedar, or XACML may reasonably ask why AEGIS does not simply use an existing policy engine for its compliance feasibility checks. The answer is that IAM and AEGIS operate at categorically different layers and answer categorically different questions.

IAM governs access decisions: given a subject, an action, and a resource, is this specific request authorized? Decisions are per-request, point-in-time, and Boolean. An ALLOW or DENY can be revised by adjusting credentials, adding permissions, or invoking exception flows. AEGIS's matroid layer governs configuration feasibility: given a proposed set of assurance properties, can this combination exist at all? The decision is structural and irreducible. A circuit violation does not depend on identity, credentials, or context; no permission grant can override a circuit, and no exception flow can compensate.

Dimension	IAM (RBAC / ABAC / OPA / Cedar)	AEGIS matroid layer
What it governs	Access decisions	Configuration feasibility
Question answered	Can subject S perform action A on resource R?	Can this combination of assurance properties coexist at all?
Decision type	Per-request, point-in-time	Structural, irreducible
Output	ALLOW / DENY	Independent (admissible) / circuit violation (BLOCKED)
Compensability	DENY overridden by grants, escalation, exception	A circuit cannot be compensated by any evidence, grant, or override
Mathematical foundation	Boolean logic; sometimes first-order	Matroid algebra (exchange axiom, circuit elimination)
Composition	Boolean ops; priority / explicit-deny-wins	Lattice meet under FAIL-dominance; non-compensatory
Role in the stack	Feeds evidence into PCUs (e.g. IAM logs $\rightarrow P_{\text{SOC2-ACCESS}}$)	Operates above the predicate layer; determines reachable configurations

The two layers are complementary, not alternative. IAM data feeds into AEGIS as evidence for predicates; IAM operates below AEGIS's predicate layer, while AEGIS's matroid layer operates above. This distinction matters operationally because IAM-only governance for AI artifacts produces an attribution gap: if an AI agent acts under a permitted IAM grant and produces a non-compliant outcome, the IAM log shows ALLOW, yet the agent's underlying configuration may have been structurally infeasible under the regulatory matroid. IAM cannot detect this, because it reasons about per-request authorization, not configuration feasibility. The expressive gap is not a matter of policy-language richness — it is a matter of mathematical object: IAM represents constraints as Boolean predicates over (subject, action, resource); AEGIS represents them as a matroid over assurance properties.

4. Formal Theorems

Theorem 1 (Soundness of Admissibility). If any hard constraint is violated, the assurance state collapses to $\perp$ and no CAI may be computed. *Proof sketch.* By construction, constraints define admissibility. Violation implies UNSAT, which maps to the absorbing element $\perp$. ■

Theorem 2 (Monotonicity Under Evidence Refinement). Let $E \leq E'$ denote refinement of evidence. If both yield admissible states, then $A(E) \subseteq A(E')$. Additional valid evidence cannot reduce assurance; it may only preserve or reveal inadmissibility. ■

Theorem 3 (Weakest-Link Compositional Safety). For composed artifacts A_1, A_2 : $A_1 = \perp \vee A_2 = \perp \Rightarrow A_1 \wedge A_2 = \perp$. System-level admissibility is bounded by the least admissible component. ■

Theorem 4 (Temporal Invalidity of Static Assurance). Any assurance state not refreshed within its validity window will either degrade or become inadmissible. Static certification is formally invalid for agentic systems [8]. ■

Theorem 5 (PCU Soundness Under Partial Evidence). For any PCU ρ evaluating predicate P with evidence bundle E : if $\rho(A, C, E) = \text{FAIL}$, then $P(A, C)$ is false. The converse is not required: ρ may return WARN when E is insufficient to decide P . False PASS is forbidden; partial telemetry yields WARN, never silent PASS. ■

5. Geometric Interpretation

AEGIS assurance is layered geometry. The CAI projection sits at the top. The assurance manifold sits in the middle, governed by lattice ordering with a metric quantifying drift on admissible regions. The constraint boundary lies below, where UNSAT collapses to $\perp$. The full assurance space sits at the base, including inadmissible regions defined by the matroid. Lattice theory governs vertical dominance. Constraint satisfaction defines the admissibility boundary. Temporal logic slices the space along the time axis [8]. The matroid defines which regions of the space are even reachable.

regions of the space along the time axis (e.g., the matroid defines which regions of the space are even reachable).

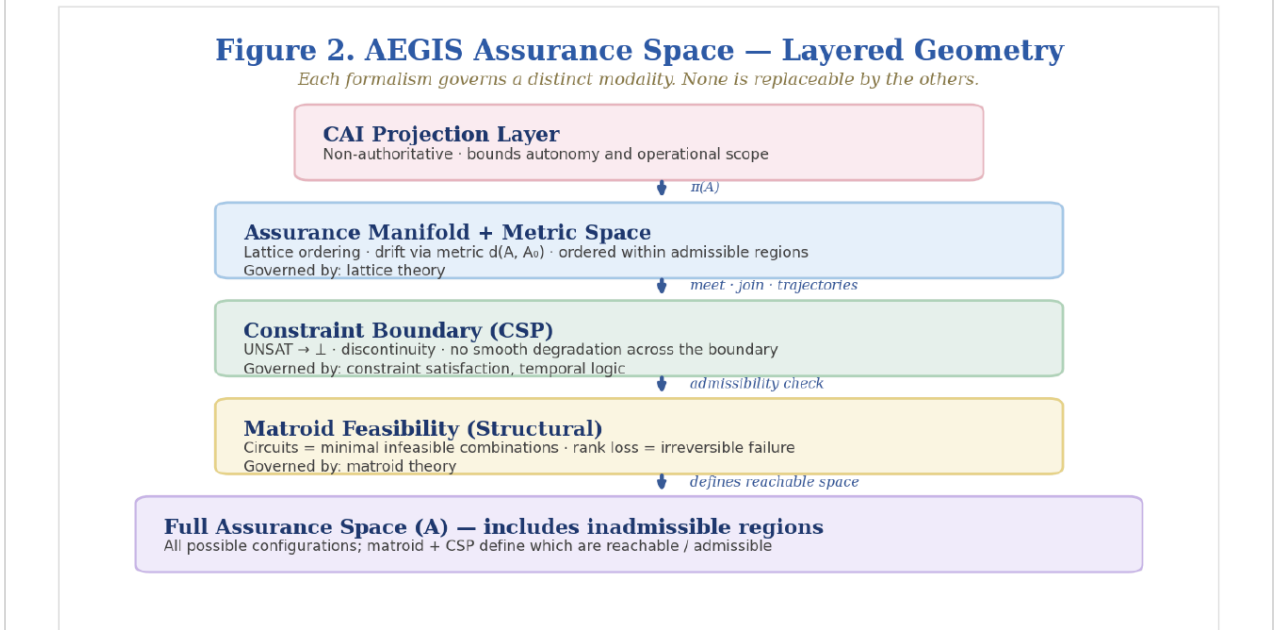


Figure 3. Layered geometry of the AEGIS assurance space. Each of the five load-bearing formalisms governs a distinct modality and is not replaceable by the others.

6. Bridging Human Trust to Assurance Dimensions

Human trust mechanisms (experience, reputation, norms, accountability) are filtered into evidence, not authority. Lattices replace holistic judgment with dominance ordering; constraints replace norms with admissibility; the matroid replaces intuition about “what can coexist” with structural feasibility; three-valued evaluation replaces vague confidence with sound verdicts under partial evidence; and temporal logic replaces memory with explicit validity windows. There exists no homomorphism from trust space to assurance space — only a filtering projection into evidence. This addresses the responsibility gap [9] at the infrastructure layer: every admissibility decision is attributable to a specific predicate evaluated against specific evidence at a specific time.

7. Multi-Domain CAI Instantiations

Medical AI. CAI bounds recommendation scope; legality and privacy constraints dominate.

Autonomous vehicles / robotics. CAI modulates speed and autonomy; stability violations collapse to safe stop.

Financial agents. CAI throttles exposure; unauthorized instruments cause immediate inadmissibility.

Infrastructure agents. CAI gates change scope; missing rollback proof blocks deployment.

Across domains: constraints decide whether, CAI decides how much.

7.5. From Compliance Controls to Predicates to PCUs

This section demonstrates how natural-language regulatory text is translated into executable AEGIS components. The pipeline is invariant across regulatory frameworks. The clause carries legal meaning. The predicate is a binary truth condition over the assurance state. The PCU computes that truth condition deterministically from observable

evidence. The gate enforces the result authoritatively. No layer is permitted to substitute for another: PCUs never interpret legal meaning, predicates never encode implementation, and gates never aggregate or score.

7.5.1 Worked Example — HIPAA §164.312(a)(2)(iv): Encryption of ePHI

Source clause. HIPAA Security Rule, 45 CFR §164.312(a)(2)(iv) — Encryption and Decryption: an addressable implementation specification directing a covered entity to implement a mechanism to encrypt and decrypt electronic protected health information (ePHI). Where a deploying entity has determined (per §164.306(d)(3)) that encryption is reasonable and appropriate — the overwhelmingly common case for ePHI at rest and in transit — AEGIS treats the resulting obligation as a hard predicate. The companion provision §164.312(e)(2)(ii) extends this to ePHI in transit. See [14] for the underlying rule.

Normalized predicate. Stripped of procedural language, the invariant the clause maintains is that no ePHI exists in an unencrypted state, whether stored or transmitted:

$$P_{\text{HIPAA-ENC}} : \forall d \in \text{ePHI}, \text{enc_at_rest}(d) \wedge \text{enc_in_transit}(d) \wedge \text{key_mgmt_compliant}(d)$$

The predicate is binary, state-based, implementation-agnostic, and non-compensatory: a single unencrypted record falsifies the universal quantifier.

Element	Specification
Evaluates	$P_{\text{HIPAA-ENC}}$
Evidence inputs	ePHI inventory, encryption state at rest, encryption state in transit, key-management attestation
Parameters	Minimum cipher strength (set by the deploying jurisdiction's standard)
Output codomain	{PASS, WARN, FAIL}
Decision rule	FAIL if any required evidence indicates non-compliance; WARN if any required evidence is missing; PASS only if all evidence is present and compliant
Side effects	None
Soundness obligation	$\text{FAIL} \Rightarrow P_{\text{HIPAA-ENC}}$ is false (Theorem 5)

Gate binding. The PCU reports to G2 (Safety) and G3 (Data Governance). Under FAIL-dominant lattice semantics, a single FAIL collapses the meet across both gates, rendering the artifact inadmissible. There is no path by which a strong showing on a fairness or transparency gate compensates for unencrypted ePHI.

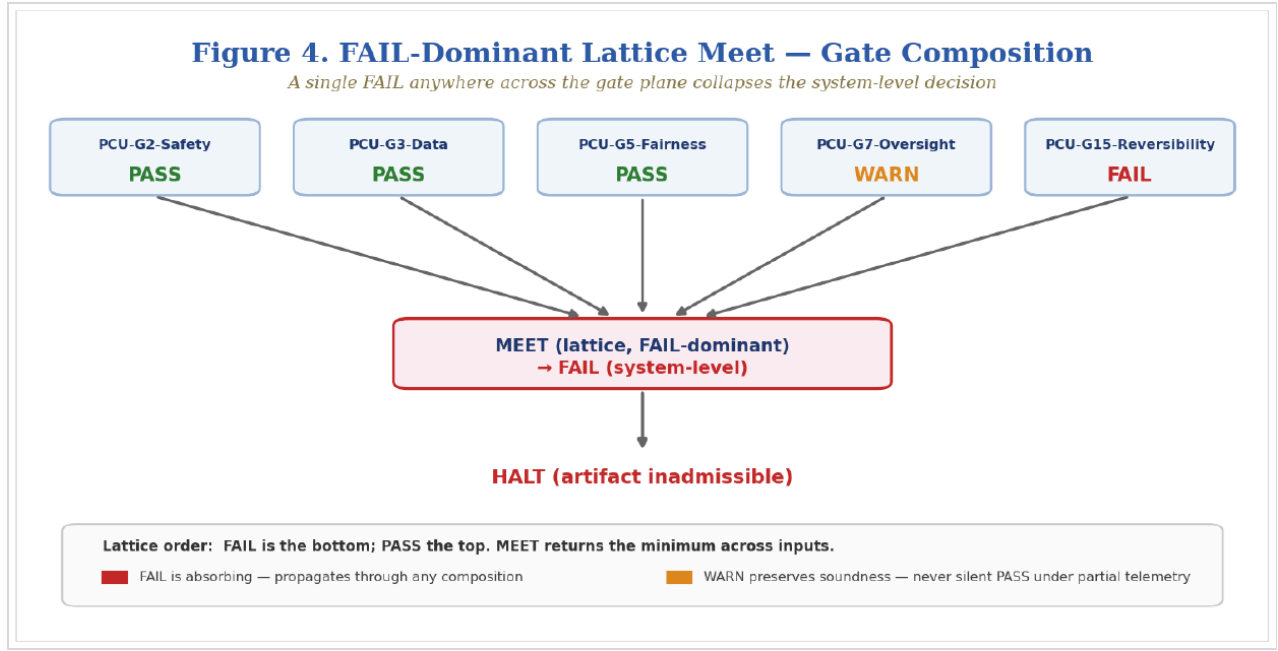


Figure 4. FAIL-dominant lattice meet. A single FAIL anywhere across the gate plane collapses the system-level decision to FAIL, irrespective of how many PASS verdicts are produced on other dimensions.

7.5.2 Structural Summary — GDPR Article 17: Right to Erasure

GDPR Article 17 [16] grants the data subject the right to obtain erasure of personal data without undue delay, with Article 12(3) bounding “undue delay” at one month, extendable by two further months (three months total) for complex or numerous requests, provided the data subject is notified within the first month. The normalized predicate is:

$$P_{\text{GDPR-ERASURE}} : \forall r \in \text{ValidRequests}, \text{deleted}(r.\text{subject}) \wedge (t_{\text{complete}}(r) - t_{\text{request}}(r)) \leq T_{\text{max}}$$

where T_{max} is a configuration parameter tied to jurisdictional context. The PCU contract follows the same structure as §7.5.1: evidence inputs are validated request records, deletion event records, and verification attestations; the decision rule returns FAIL on any overdue or unverified deletion, WARN on missing telemetry, and PASS otherwise. The gate binding is G15 (Reversibility & Recovery) with secondary binding to G3 (Data Governance).

7.5.3 Structural Summary — SOC 2 CC6.1: Logical Access

SOC 2 Common Criterion CC6.1 [15] requires logical access controls over protected information assets. In the exact AICPA wording, the criterion states that the entity implements logical access security software, infrastructure, and architectures over protected information assets to protect them from security events to meet the entity's objectives. The underlying state invariant is:

$$P_{\text{SOC2-ACCESS}} : \forall e \in \text{AccessEvents}, \text{auth_strength}(e) \geq \text{required}(e.\text{role}) \wedge e.\text{subject} \in \text{authorized}(e.\text{resource}, e.t) \wedge \text{role_freshness}(e.\text{subject}) \leq \tau$$

Specific implementation choices (multi-factor authentication, role-based access control, periodic access reviews) enter the PCU as evidence sources, not as authoritative requirements; the AICPA points of focus are guidance, not mandatory controls. The gate binding is G2 (Safety) with secondary binding to G6 (Audit & Transparency).

7.5.4 What the Examples Establish

First, legal text and execution code never touch directly. The predicate layer mediates between them, allowing AEGIS to remain stable when implementations change while keeping the regulatory binding intact. **Second, a single predicate frequently satisfies multiple frameworks.** The encryption predicate in §7.5.1 carries satisfaction value for security-of-processing requirements across multiple regulatory regimes; the access predicate in §7.5.3 likewise. **Third, PCUs are sound but may be incomplete.** Theorem 5 formalizes this: false PASS is forbidden; partial telemetry yields WARN, never silent PASS. This is what makes the framework defensible to auditors — a regulator can verify the soundness obligation once and rely on PCU output thereafter.

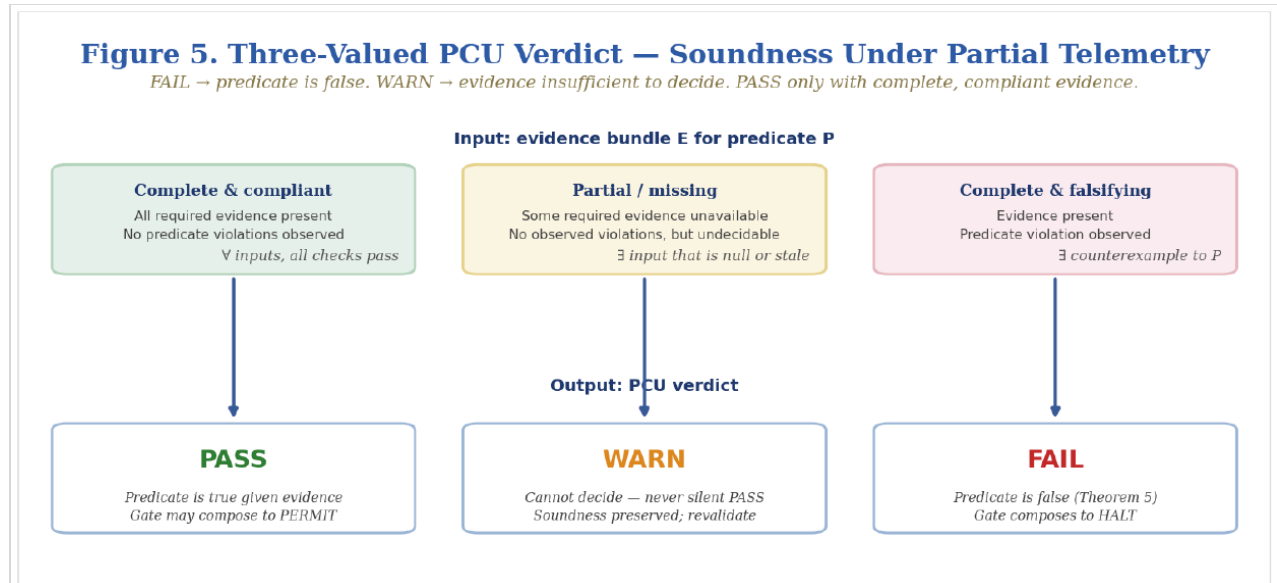


Figure 5. Three-valued PCU verdict preserves soundness under partial telemetry. A boolean evaluator collapses the middle column into the left one; a scoring evaluator averages all three. AEGIS preserves the distinction.

Together, these properties establish that compliance under AEGIS is provable rather than declarative.

8. Related Work

AEGIS sits adjacent to a fast-developing literature on runtime governance for agentic AI, but is distinguished by its mathematical foundations. **Score-based runtime governance.** Governance-as-a-Service [1] and Charli AI's Governance Control Plane [2] compute Trust Factors with severity-aware penalization; AEGIS rejects scoring as authoritative, treating any score as a non-authoritative projection (the CAI) and reserving authority for lattice-based admissibility decisions. **OPA-style declarative rule enforcement.** Governance-Aware Agent Telemetry [3] uses OPA-compatible rules for sub-200ms enforcement; AEGIS's predicate layer subsumes the OPA rule layer but adds three-valued semantics, the soundness obligation of Theorem 5, and the matroidal infeasibility check that no OPA rule set can express. **FSM-based runtime conformance.** MI9 [4] uses finite-state-machine sequence matching; AEGIS's gate plane is finer-grained and composable under FAIL-dominant lattice meet. **Kubernetes-native AI RMF control planes.** AAGATE [5] operationalizes the NIST AI RMF; AEGIS is a foundational algebra rather than a Kubernetes architecture, intended as the formal substrate against which AAGATE-style systems can be evaluated. **Layered translation methods.** Recent work [10] proposes layered translation from governance norms to enforceable controls; AEGIS provides the mathematical structure such layering presupposes.

9. Conclusion

AEGIS Algebra replaces subjective trust with executable assurance. By integrating five load-bearing mathematical formalisms into a single structured space—matroid feasibility, lattice dominance, constraint-based admissibility, three-valued soundness, and temporal validity—it enables bounded autonomy, compositional safety, temporal validity, and regulator-grade auditability for agentic AI systems. This is not a policy framework or scoring system; it is an algebra for governing autonomous behavior. The framework is presented openly; its instantiation is realized through a control-plane implementation and a demonstration system, backed by a proprietary engineering corpus that operationalizes the theory against specific regulatory regimes and evidence ecosystems.

Beyond individual artifacts, future AI ecosystems will increasingly consist of interacting agents, tools, models, organizations, and regulatory authorities operating as distributed socio-technical systems. While the present work focuses on executable assurance for individual artifacts, the algebra naturally suggests higher-order extensions. The categorical component (Assure) provides a foundation for reasoning about assurance-preserving composition of agents and workflows; the metric and temporal components provide a basis for feedback-driven assurance maintenance under system drift; and the assurance state itself can be extended to incorporate provenance, delegation, and epistemic accountability across multiple actors. These extensions motivate a broader notion of compositional assurance, in which admissibility becomes a property not only of individual artifacts but of the transformations, interactions, and governance structures connecting them.

More generally, executable assurance may be viewed as a foundational layer beneath executable governance. The formalisms presented here establish structural feasibility, admissibility, sound evaluation under partial evidence, and temporal validity for autonomous systems. Future work may extend this foundation with control-theoretic models of adaptive assurance, epistemic formalisms for knowledge and evidence provenance, graph-based representations of delegation and accountability relationships, and mechanism-design approaches for incentive-compatible governance among autonomous agents. Such extensions would preserve the non-compensatory and deterministic character of AEGIS while enabling governance of increasingly complex multi-agent ecosystems and AI-mediated institutions. “In this view, AEGIS becomes not merely an assurance algebra for individual AI artifacts, but a candidate mathematical substrate for governing distributed societies of humans, agents, tools, and autonomous institutions operating under bounded autonomy.”

Future work includes extending the matroidal layer to parametric and online matroids for adaptive systems, and strengthening the temporal-dynamics treatment of assurance decay between revalidation events.

Acknowledgments

The control-plane implementation and demonstration system at Inferloop Technologies, which instantiate this algebra in practice, informed the framework’s design — particularly the matroidal treatment in §3A and the worked examples in §7.5.

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Appendix I — Matroid Theory and Structural Feasibility

This appendix formalizes the matroidal foundations underlying AEGIS Algebra. Let E be the finite set of assurance parameters. A matroid $M = (E, \mathcal{I})$ defines which subsets of parameters are structurally feasible. Independence corresponds to co-realizable assurance properties; dependence encodes irreducible incompatibilities. Circuits represent minimal infeasible combinations. When a circuit is activated through system evolution or configuration, feasibility collapses irreversibly, explaining abrupt assurance failure without recourse to scoring. Rank corresponds to maximal realizable assurance dimensionality. Rank loss explains why systems may permanently lose autonomy class even after partial remediation. Extensions to parametric matroids and online matroid learning are reserved for future work.

Appendix II — Lattice Ordering and Absorbing Failure

Given matroid-feasible states, assurance states form a partially ordered lattice $(\mathcal{A}, \subseteq)$. The meet operator corresponds to weakest admissible behavior; the bottom element $\perp$ is absorbing. Any operation involving $\perp$ yields $\perp$. This ensures failure cannot be masked by aggregation or averaging.

Appendix III — Constraint Satisfaction and UNSAT Semantics

Constraints encode regulatory, legal, and safety requirements as non-negotiable predicates. The assurance state is admissible iff all constraints are satisfied. UNSAT states collapse to $\perp$ regardless of lattice position or metric proximity. This appendix clarifies the distinction between infeasibility (matroidal) and inadmissibility (constraint violation): the former concerns whether a configuration can exist at all; the latter concerns whether an existing configuration satisfies its hard constraints.

Appendix IV — Temporal Logic and Validity

Assurance states are time-indexed. Temporal logic enforces revalidation and decay: an assurance state carries a validity window, and a state not refreshed within that window degrades or becomes inadmissible. Static assurance is formally invalid for agentic systems; this is the formal content of Theorem 4.

Appendix V — Mapping Human Trust to Assurance Evidence

Human trust mechanisms (reputation, norms, experience) are mapped into evidence channels, never decision authority. There exists no homomorphism from trust space to assurance space — only filtered projection into evidence. Every admissibility decision therefore remains attributable to a specific predicate, evaluated against specific evidence, at a specific time.

Appendix VI — Reference Architecture

A reference AEGIS implementation comprises a parameter schema (the matroid ground set E), a constraint engine (CSP/SAT layer), a lattice evaluator, a temporal validity loop, and the CAI projection module. Predicate Compute Units sit between the constraint engine and the gate layer, evaluating individual predicates against evidence and emitting structured verdicts. Gates compose verdicts under FAIL-dominant lattice meet to produce authoritative control actions. Specific predicate sets, PCU implementations, evidence-source connectors, and gate orderings constitute the deployment corpus and are governed by separate engineering documentation.

Appendix VII — Research and Regulatory Positioning

AEGIS differs fundamentally from scoring frameworks, LLM-based safety tools, OPA-style policy engines, and static audits. It is a mathematically grounded control system for autonomy, defined by the integration of five load-bearing formalisms—matroid theory, lattice and order theory, constraint satisfaction, three-valued evaluation logic, and temporal logic—into a single algebra. This positioning makes it suitable for academic citation, regulatory adoption, and long-term extensibility, while preserving the operational corpus required for deployment as a separate proprietary asset.

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